



9.5.1 Additional homework ¶

Homework 9.5.1.1. Let $\|\cdot\|$ be matrix norm induced by a vector norm $\|\cdot\|$. Prove that for any $A \in \mathbb{C}^{m \times m}$, the spectral radius, $\rho(A)$ satisfies $\rho(A) \leq \|A\|$.

Some results in linear algebra depend on there existing a consistent matrix norm $\|\cdot\|$ such that $\|A\| < 1$. The following exercise implies that one can alternatively show that the spectral radius is bounded by one: $\rho(A) < 1$.

Homework 9.5.1.2. Given a matrix $A \in \mathbb{C}^{m \times m}$ and $\epsilon > 0$, there exists a consistent matrix norm $\|\cdot\|$ such that $\|A\| \leq \rho(A) + \epsilon$.